higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: GENERAL

1.1 1.1.1 True
1.1.2 False
1.1.3 False
1.1.4 True
1.1.5 False

(5 × 1) (5)

1.2 1.2.1 A
1.2.2 B
1.2.3 D
1.2.4 C
1.2.5 A

(5 × 1) (5)
[10]

QUESTION 2: PLASTERING

2.1 2.1.1 Remove all dust, dirt and oily substances from the walls.

2.1.2 Remove all protrusions.

2.1.3 Check that the substrate is plane and vertical.

2.1.4 If necessary, roughen the substrate to allow the plaster to bond properly with it.

2.1.5 Make sure that the wall is strong enough.

2.1.6 Test the area for water absorbency.

2.1.7 On the day before plastering, wet the area well.

2.1.8 Clear the surrounding area of all materials and tools that you do not need for the job.

(8 × 1) (8)

2.2
- Spade
- Bucket
- Wheelbarrow
- Wooden hawk
- Steel hawk
- Plastering trowel
- Angle tool
- Square and bevel
- Straight edge

(Any 3 × 1) (3)

2.3 Spread mortar onto the brickwork.✓ Wait for the mortar to dry until it is thumbprint hard.✓ Check the mortar area for places where it needs to be filled in or smoothed.✓ Use a short square jointer to rake mortar from the joints.✓ Create lines that are perpendicular to the brickwork and have no curve to them.✓ Strike the vertical joints first,✓ after striking the joints, softly brush them to remove any loose mortar✓ and remove excess mortar from the surround.✓ (8)

2.4 Apply surface retarder instead of shutter-release oil to shutters and then wash or brush off the retarded outer layer of paste before it hardens.✓ Strip the formwork early and wire brush the concrete surface. Apply a spatter-dash coat,✓ abrasive blasting using water or oil-free compressed air.✓ Hack or bush-hammer smooth the concrete surface.✓ Attach the expanded metal using power-actuated nails (Hilti gun) and large fender washers.✓ (5)

2.5 It should be 10✓–16 mm thick.✓ (2)

2.6
- At the edge of a damp course (DPC)
- Next to columns
- Where the substrate has been laid in sections (3)

2.7 It is the resulting concrete structure once the shuttering or formwork has been removed (1)

[30]

QUESTION 3: SCREEDS AND TOPPINGS

3.1 Before you can lay a screed or topping, the slab that is going to form the substrate must be ready.✓ If not, the screed or topping will not bond to the slab,✓ and it will separate and crumble.✓ (3)

3.2 Monolithic screeds and toppings✓ which are applied to the surface while the concrete in the base is still in a plastic state.✓ Bonded screed and toppings✓ which are applied to hardened concrete floors✓ and unbonded screeds and toppings✓ which are used when it is impossible to ensure a good bond between floor and screed or topping, in this case the screed and or topping is separated from the floor by insulation boards.✓ (6)

3.3
- To depress large aggregates below the surface.
- To remove slight imperfections, lumps and voids.
- To produce a level or plane surface.
- To compact the finish and consolidate mortar at the surface in preparation for other finishing operations. (4)

3.4 To provide fixing points (1)

3.5 A colour hardener is a material that strengthens and colours concrete floors.✓✓ We use it because it is very hard-wearing, has exceptional abrasion resistance and reduces drying shrinkage.✓✓ (4)

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3.6 Joints must be cut to a depth one-quarter of the topping thickness. (1)

3.7 Use a strong detergent✓ and boiling water to help remove grease.✓ (2)

3.8 Work a small area at a time.✓ Slush the grout mix over the area where it is needed.✓ Brush the grout in manually, using a stiff nylon scrubbing brush.✓ Use a brush with bristles of 30–50 mm long, so that they are flexible enough to reach into any depressions. Brush the grout out again, leaving only a very thin coating on the surface with no pools of grouting in the depressions.✓ Lay the screed within 10 to 20 minutes.✓ (5)

3.9 An intercoat is a rich layer of sand, cement✓ and water mix used to form a bonding layer between the base slab and the screed or topping✓ (2)

3.10
- Portland cement
- Rapid-hardening cement
- Portland blast furnace cement
- Sulphate-resisting Portland cement
- Special Portland cement (Any 2 × 1) (2)

[30]

QUESTION 4: TILING

4.1 4.1.1 Granolithic floors are used for large factory floors where heavy loads are left for storage.

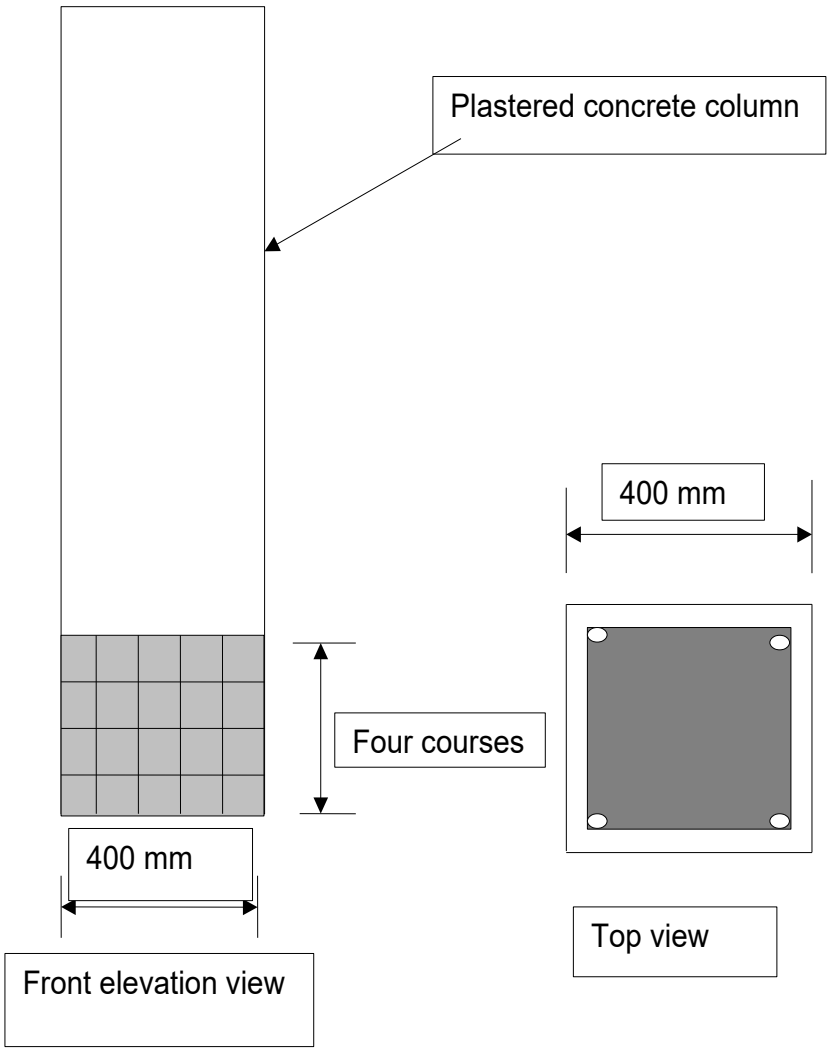
4.1.2 PVC tiles are very popular in school buildings, hospitals, public buildings and offices.

4.1.3 Thermoplastic tiles are used in areas where there is no grease and oil such as in offices and schools.

(3 × 1) (3)

4.2 The paper lining, which is the 'skin' of the plasterboard, is unsuitable for the application of tiles,✓ as the moisture in the adhesive will cause the paper to debond from the plaster-board.✓ This will cause the tiles to fall off✓ and the plaster-board will begin to lose its board-like structure.✓ (4)

4.3



Correctness of views	9
Dimensions	2

(13)

4.4 We use a 2 m long straight edge to check the tolerance of the finished floor.✓ Start on one side of room or tiled area. Select a suitable piece of hardboard or other material to make two 'feet' 3 mm thick to support the straight edge.✓ Placing one 3 mm thick 'foot' under each end of the straight edge, lay it across a section of tiling.✓ If the tiles obstruct (block) the straight edge at any point, they are more than 3 mm above the required level.✓ If the gap between the bottom of the straight edge and the tiles is more than 6 mm at any point, that point is more than 3 mm below the required level.✓ (5)

4.5 Make sure that the joints between the tiles are clean and free of any debris.✓ Use a float to spread the grout mixture across a small area of the tiled floor.✓ Hold the float at an angle so as to push the grout into the grout joints.✓ Grout will get onto the tiles. This is in order. You can clean it off after the grout sets.✓ (4)

4.6 Periodically tap the surface of the laid tiles with a hammer. If you hear a hollow sound, this means that the tile adhesive is not correctly distributed over the under-surface of the tile. (1)

[30]

TOTAL: 100